

Tharinda Abeywardhana

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🌐 Portfolio

🌐 LinkedIn

🐙 GitHub

I like understanding why things work the way they do, then building them to find out if I'm right. Most of my projects start with a question, why does this converter ring at that frequency, why is this signal noisier than it should be, and end with a PCB and a scope probe. I've spent the last couple of years working across power electronics, embedded systems, and high-speed PCB design, and what I enjoy most is the point where the schematic meets real hardware and you have to figure out what you got wrong. I'm looking for a hardware research internship where I can work on real problems alongside people who care about getting the details right.

EDUCATION

University of Moratuwa, Sri Lanka

March 2024 – Present

- B.Sc. Engineering (Hons.) in Electronic and Telecommunication Engineering
- **Cumulative GPA (end of 4th Semester):** 3.74 / 4.00 (93.5%) [Transcript]
- Dean's List: Semester 1

Dharmapala Vidyalaya, Colombo, Sri Lanka

- GCE Advanced Level (2022), Physical Science Stream, Combined Mathematics (A*), Physics (A*), Chemistry (A*). Top 1.2% nationally.

Certifications

- Power Electronics Specialization CU Boulder (Prof. R. Erickson)
- Introduction to FPGA Design for Embedded Systems CU Boulder
- Hardware Description Languages for FPGA Design CU Boulder
- Differential Equations for Engineers HKUST
- System Verilog for ASIC/FPGA Design & Simulation Dept. of ENTC, University of Moratuwa

TECHNICAL STRENGTHS

Circuit Simulation & Analysis	LTspice, NI Multisim, Altair PSIM, MATLAB
PCB Design & Layout	Altium Designer (multi-layer, high-speed, power distribution)
Hardware Description Languages	SystemVerilog, Verilog, VHDL
FPGA & Digital Design	Xilinx Vivado, Intel Quartus, Verilator
Embedded Systems	STM32, ESP32, Raspberry Pi, ROS 2
Programming Languages	C/C++ (embedded), Python, MATLAB, Java
RF & Signal Processing	GNU Radio, bladeRF, MATLAB Signal Processing
Mechanical Design	SOLIDWORKS

HARDWARE PROJECTS

Autonomous Indoor Drone Swarm Mapping System

Jun 2026 to Present

- Building a swarm of 5 to 6 custom mini quadcopters for collaborative indoor mapping, with custom PCBs integrating IMU sensors, motor drivers, power management, and radio.
- Developing flight-control firmware from scratch on STM32, including sensor fusion, attitude estimation, and cascaded PID stabilization.
- Implementing real-time localization and occupancy-grid mapping for GPS-denied indoor environments.
- Implementing inter-drone communication and multi-agent coordination for exploration, coverage, and collision avoidance.

Smart Autonomous Floor Mopping Robot

Feb 2026 to Jul 2026

- Served as system architect, power electronics lead, and autonomous navigation engineer in a team of five building a fully autonomous floor cleaning robot.
- Designed a power management PCB with LiPo charge/discharge control and three regulated rails (3.3 V, 5 V, 12 V) with per-rail current monitoring, budgeted for 3+ hours of continuous runtime across three subsystems.
- Designed a sensor management PCB around an STM32 for cliff and obstacle detection, wheel encoder interfacing, pump control, and closed-loop motion feedback, and wrote the firmware for both boards, battery state monitoring, charge/discharge state machines, fault protection, and inter-board communication.
- Led the autonomous navigation stack using LiDAR-based SLAM, path planning, and coverage navigation on Raspberry Pi with ROS 2.

Tools: Altium Designer, STM32, Embedded C, LTspice, ROS 2, LiDAR, Raspberry Pi

Repository:  GitHub

Advanced 12-Layer High-Speed PCB System Design, iMX6 Rex

Jan 2026 to May 2026

- Designed a 12-layer PCB from scratch based on the open-source FEDEVEL iMX6 Rex schematic (NXP i.MX6, 1.2 GHz quad-core, DDR3-1066), with a Signal/GND/Signal/GND/Power×4/GND/Signal/GND/Signal stackup optimized for signal integrity and low-impedance power delivery.
- Routed length-matched, impedance-controlled differential pairs for DDR3, PCIe, SATA, Gigabit Ethernet, HDMI, LVDS, and USB with controlled return paths.
- Performed BGA fanout/escape routing and designed a multi-rail PDN (1V375, 1V5, 2V5, 3V3) with hierarchical decoupling to minimize voltage droop.
- Applied EMI/EMC reduction techniques, thermal-aware component placement, and DFM/DRC validation in Altium Designer.

Tools: Altium Designer **Repository:**  GitHub

Buck Converter Design, MIT 6.622 Specification

Mar 2026 to May 2026

- Designed a high-efficiency synchronous buck converter (24 to 36 V input, 14 V output, 50 to 125 W) to the MIT 6.622 specification, switching at 100 kHz.
- Designed a Type III analog compensator in MATLAB achieving a **+50.96° phase margin** and infinite gain margin, after analytically ruling out Type I and Type II for the LC-resonant plant ($f_{LC} \approx 1.79$ kHz).
- Verified closed-loop regulation in Altair PSIM across a 20–40 V input sweep, holding output at 13.84 V with ripple under 100 mV.
- Achieved **92% simulated efficiency** at 28 V against a 90% requirement, with a full loss budget down to 3.5% total losses and junction temperatures within limits (MOSFET $\approx 76^\circ\text{C}$, diode $\approx 104^\circ\text{C}$) across -20 to $+50^\circ\text{C}$ ambient.

Tools: LTspice, MATLAB, Altair PSIM, Altium Designer **Repository:**  GitHub

ThermoBand, Wearable Temperature Monitoring Device

Feb 2025 to Jul 2025

- Designed two custom 4-layer PCBs for a wearable temperature monitoring system with wireless telemetry.
- Developed the power management PCB with LiPo charge/discharge control, integrated voltage regulation, multi-rail distribution, and protection circuitry, achieving **12+ hours** of continuous battery operation.
- Designed the sensor interface PCB with precision analog conditioning (amplification and low-pass filtering) reaching $\pm 0.32^\circ\text{C}$ resolution, with on-board ESP32 telemetry and a signal path optimized for noise rejection in a wearable form factor.

Tools: Altium Designer, Solidworks **Repository:**  GitHub

Smart IV Fluid Monitoring and Alert System

Jul 2025 to Dec 2025

- Designed a custom 2-layer PCB integrating an ESP32 microcontroller, IR sensing front-end, and Wi-Fi telemetry for real-time IV fluid monitoring.
- Implemented an analog sensor conditioning stage for IR signal processing with amplification and filtering, resolving drip rates down to **1 drop per minute** for accurate low-flow detection.
- Developed the system for continuous monitoring with real-time wireless alerting on low-flow and empty-bag conditions.
- Managed power delivery constraints and battery efficiency to support reliable continuous operation.

Tools: Altium Designer, ESP32 Firmware, Solidworks

Repository:  GitHub

Fully Analog Acoustic Guitar Tuner

Jul 2025 to Dec 2025

- Designed a fully analog guitar tuning device with no digital signal processing.
- Developed the analog signal-conditioning chain including a low-noise microphone preamplifier and precision band-pass filters covering all six strings at their fundamental frequencies.
- Implemented a frequency-to-voltage conversion stage for tuning indication, achieving ± 2 Hz accuracy, validated by tuning a real guitar.
- Designed a shielded enclosure with proper grounding and EMI mitigation to preserve front-end signal integrity.

Tools: LTspice, Altium Designer, Solidworks

Repository:  GitHub

ADDITIONAL ENGINEERING PROJECTS

Two-Way Digital Paging System - Software Defined Radio

Jul 2025 to Dec 2025

- Designed and implemented a complete two-way communication system using SDR hardware (*bladeRF*) instead of fixed radio transceivers.
- Built the full baseband signal processing chain including audio conditioning, packet framing with addressing, acknowledgment handling, and digital modulation using BPSK/QPSK.
- Deployed the system on the GNU Radio framework for end-to-end implementation and testing.
- Validated system operation through real-world RF experiments.

Tools: GNU Radio, bladeRF, MATLAB

Repository:  GitHub

VOLUNTEER AND LEADERSHIP EXPERIENCE

Electronic and Telecommunication Engineering E-Club, University of Moratuwa

- **HR Coordinator and Event Organizer (Eid26)** Aug 2025 – Present

Sasnaka Sansada Foundation

- **LearnSteer National Project Coordinator** Dec 2024 – Nov 2025
Led nationwide Physics workshops and examinations for GCE A/L students.
- **Member, Colombo District Team** Jun 2023 – Present
Educational outreach and teaching support for rural school students.

REFERENCES

Dr. Ranga Rodrigo

Senior Lecturer

Department of Electronic and Telecommunication Engineering

University of Moratuwa, Sri Lanka

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Dr. Upeka Premaratne

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